

PPA-4: Symmetry-Governed Computation for Physical and Biological Systems

Title

Symmetry-Constrained Computational Methods for Physical, Chemical, and Biological Systems

Field of the Invention

The invention relates to computational modeling and implementation of physical, chemical, and biological systems, particularly systems whose state evolution is governed by symmetry constraints rather than arbitrary symbolic rules.

Background

Many physical systems exhibit:

- discrete allowed states,
- forbidden transitions,
- error resistance via structural constraints.

Conventional computation struggles to model or implement such systems faithfully because Boolean logic allows arbitrary state transitions.

Summary

The invention discloses computational methods in which:

- allowed state transitions are defined by symmetry,
- illegal transitions are structurally impossible,
- and computation proceeds via constrained spatial evolution.

These methods may be used to model or implement:

- genetic translation,

- molecular folding,
 - crystallization,
 - learning systems,
 - and other symmetry-governed processes.
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Detailed Description

Symmetry-Constrained State Evolution

States exist only within symmetry-allowed configurations.
Transitions violating symmetry do not exist.

Error Prevention by Structure

Unlike error-correcting codes, the system prevents errors by eliminating illegal states entirely.

Computational Interpretation

Such systems perform computation by:

- exploring allowed state paths,
 - converging toward stable configurations,
 - without symbolic programs.
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Advantages

- Physically realistic
 - Robust to noise
 - Naturally parallel
 - Suitable for modeling life-like systems
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Provisional Statement

This application claims priority for symmetry-governed computational methods applicable to physical and biological systems.